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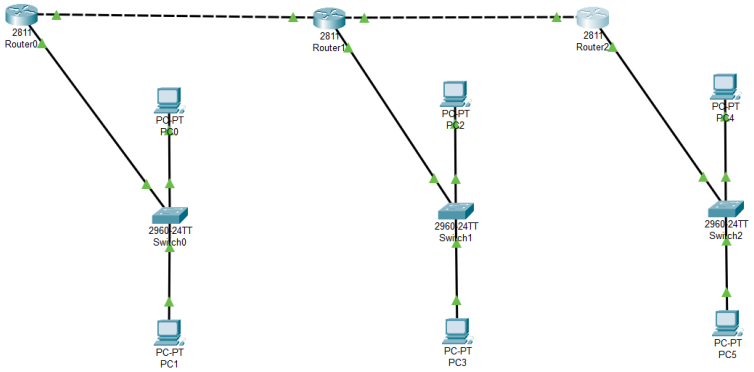

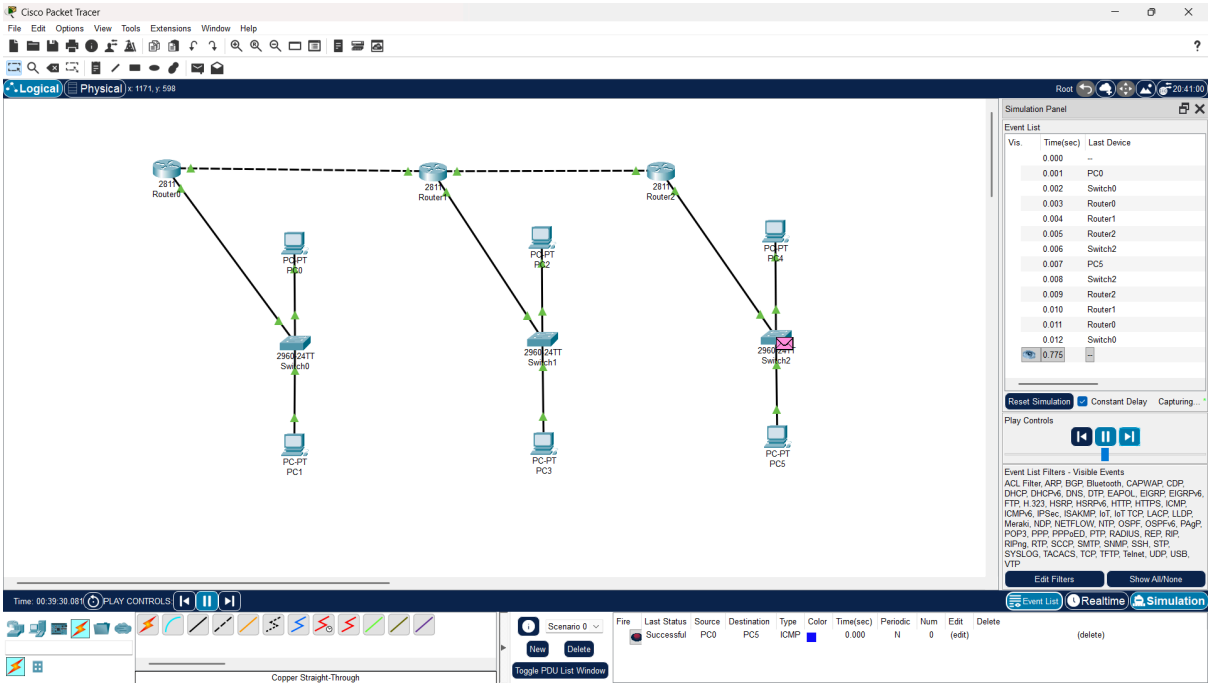

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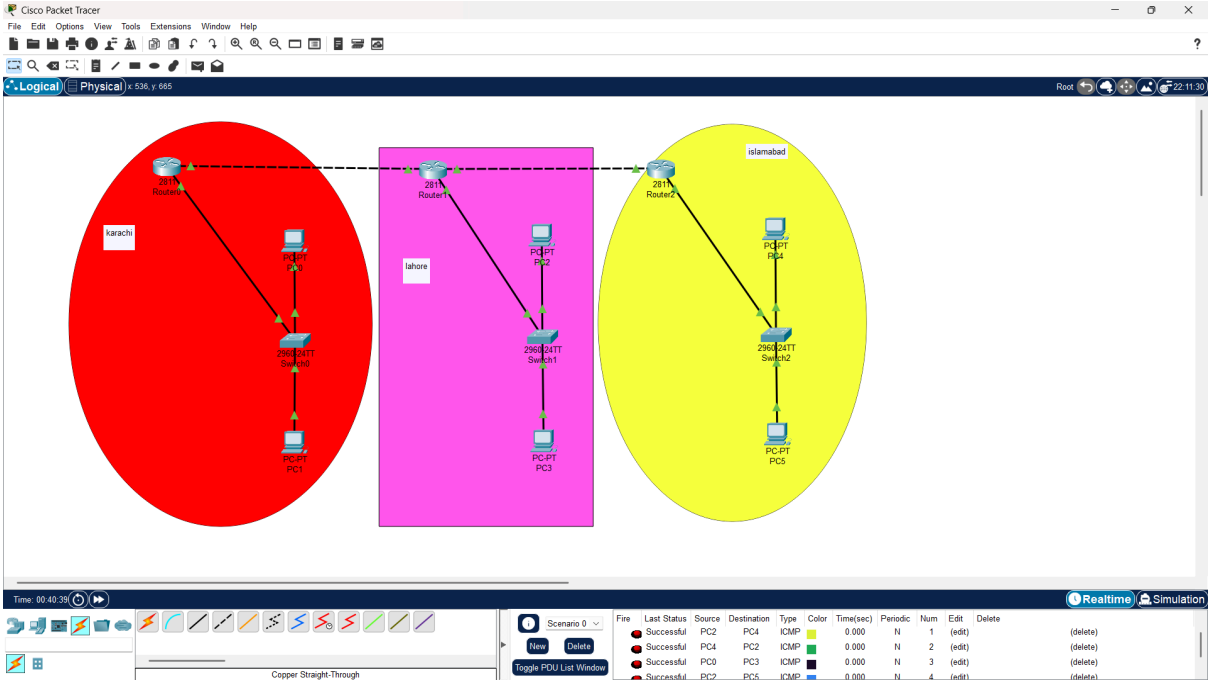

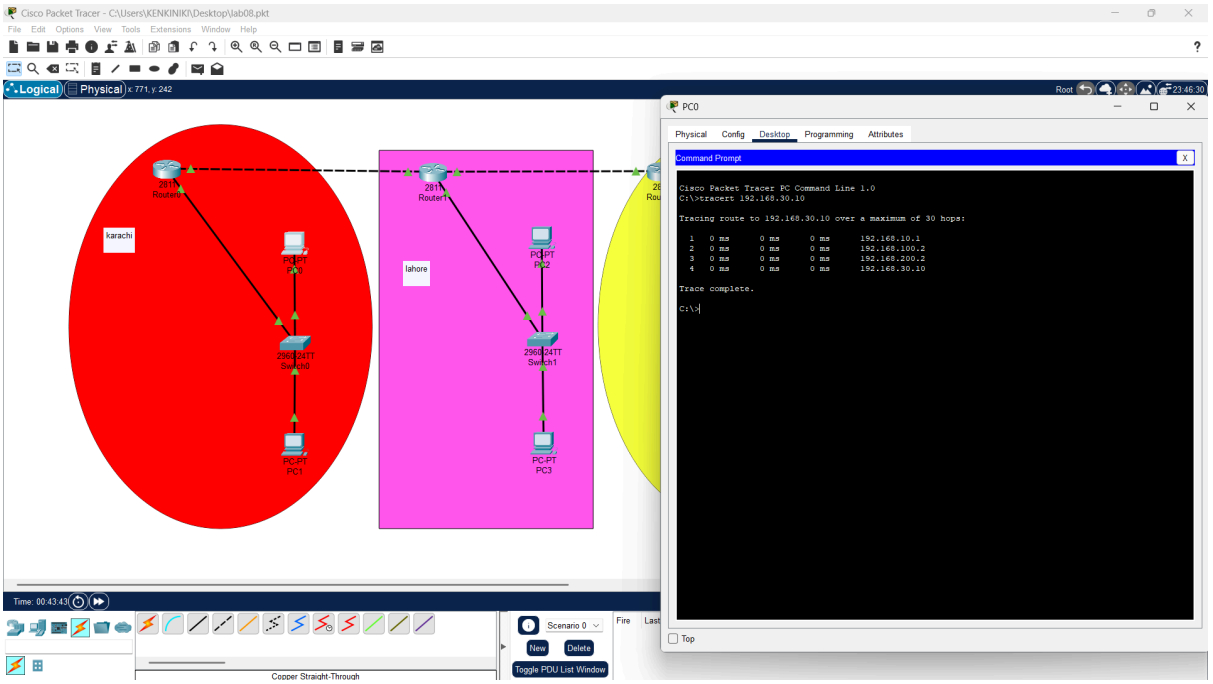

Summary Table (for all routers)

Office	Interface	IP Address	Connects To
Karachi	Fa0/0	192.168.10.1	Karachi LAN (Switch)
	Fa0/1	192.168.100.1	Lahore (Fa0/1)
Lahore	Fa1/0	192.168.20.1	Lahore LAN

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	Fa0/0	192.168.100.2	Karachi (Fa0/1)
	Fa0/1	192.168.200.1	Islamabad (Fa0/1)
Islamabad	Fa0/1	192.168.30.1	Islamabad LAN
	Fa0/0	192.168.200.2	Lahore (Fa1/0)

DIAGRAM

```
[Karachi LAN]        [Lahore LAN]        [Islamabad LAN]
  PC0                 PC2                  PC4
 192.168.10.10        192.168.20.10        192.168.30.10
    |                  |                    |
   SW_Karachi          SW_Lahore            SW_Islamabad
    |                  |                    |
 R_Karachi -------- R_Lahore --------- R_Islamabad
 Fa0/0:192.168.100.1  Fa0/1:192.168.100.2  Fa0/0:192.168.200.2
 Fa0/1:192.168.10.1    Fa1/0:192.168.20.1    Fa0/1:192.168.30.1
                       Fa0/0:192.168.200.1
```

CONFIGURATION CODE

```
enable
configure terminal
interface FastEthernet0/1
 no shutdown
end
```

For giving hosts: hostname R_Islamabad

To check interfaces link of Routers: show ip interface brief

--- Static Routes ---
ip route 192.168.20.0 255.255.255.0 192.168.200.1
ip route 192.168.10.0 255.255.255.0 192.168.200.1

Answers

Why Lahore router requires different routing logic than the other two?
Lahore is the central hub — it has two router-to-router links (to Karachi and Islamabad) and its own LAN, so it must have explicit routes and active interfaces for both directions and be able to forward traffic between the two branches. Karachi and Islamabad only have a single

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neighbor (Lahore), so each simply points all non-local traffic to Lahore; they don't need to forward between two remote sites. Thus Lahore needs two static routes (one to each remote LAN) and must be able to act as the next-hop for traffic passing between Karachi and Islamabad.

Why static routing for this assignment?
Static routing is simple, deterministic, low-overhead, and cost-effective for small, fixed topologies — appropriate for a three-site lab. It requires manual updates if networks change, which is acceptable here.

Limitations & recommendations for production
Static routing scales poorly as you add sites. For production or when links change, use a dynamic protocol (OSPF/EIGRP) and add redundancy (dual links, HSRP/VRRP for gateway redundancy). Add ACLs and management access control for security.